

Anonymity as an Aggregation Condition:

Governance, Resource Structure, and Membership Dynamics in Twelve-Step Mutual-Aid Organizations

Working paper. Author anonymous by design; see the scope note.

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Abstract

Alcoholics Anonymous (AA) has operated for nearly ninety years across approximately 120,000 autonomous groups under a written constitution, the Twelve Traditions, that has never been substantially revised. This paper develops three connected formal results and reports the robustness analysis that determines how much each of them is worth.

First, and centrally, three of the Traditions (group conscience with servant leadership, the refusal to organize hierarchically, and anonymity with principles placed before personalities) are read as jointly enforcing the condition that maximum influence weight within a group vanishes as the group grows, which Golub and Jackson (2010) prove is necessary and sufficient for naive opinion averaging to aggregate dispersed private information. If the reading is right, AA's governance rules implement a known aggregation criterion, reached inductively in 1946, six decades before its formal statement. *The reading is the paper's largest unverified step and is flagged as such wherever it is used.* Two corollaries not stated in the Traditions follow: rotation of service satisfies the criterion only if the rotation pool scales with group size, and whether a concentration of attention obstructs learning at all depends on how it scales with the group rather than on how severe it looks at any one size.

Second, we derive the coupling between the Twelve Steps and Twelve Traditions through an intermediate layer of group-produced resources, rejecting the natural conjecture that the two lists pair by index and recovering, without assuming them, a two-tier functional division among the Traditions and the primacy of unity that Tradition 1 asserts of itself. We then report, against our own interest, that five of the twelve rejections are structurally forced by sparsity and invisible to multiplicative perturbation, that a threshold design able to see them finds them failing by comparable margins, and that the central coupling claim survives structural randomization in only 40.6 per cent of draws, which is less often than it fails. The coupling results rest on the magnitudes of two hand-written matrices and are argued for rather than certified.

Third, a stochastic simulation with endogenous membership, respecting the institutional fact that an AA group cannot refuse *membership*, distinguishes two channels of newcomer supply and shows that groups starved of exogenous referrals decline demographically while per-member quality among survivors is maintained or rises, whereas groups with exclusionary internal culture shrink without dissolving. Sensitivity analysis over all 118 hand-chosen parameters, global and one-at-a-time, plus Morris screening, Sobol indices, four structural variants and a structural randomization, finds that exactly two simulation results survive: that a fully adherent group persists, and that losing referrals is worse than losing attraction *in the sense of being more likely to kill the group*. Read as “ends up smaller” that ordering reverses under three of four architectural changes. No parameter is estimated. The contribution is a correspondence, a derivation, an accounting of what the analysis cannot support, and a set of falsifiable predictions, several testable with records AA service structures already keep.

Scope and ethics. This paper does not reproduce the text of the Twelve Steps or Twelve Traditions, which is copyrighted by Alcoholics Anonymous World Services, Inc.; short paraphrases are used throughout. AA is not affiliated with this work and, by its Sixth Tradition, could not be. Nothing in this paper can assess any individual's recovery, and it

must not be used to do so. No AA publication was acquired for this project; the historical material is drawn from independent scholarship and from primary temperance sources in the public domain.

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1. Introduction

Two literatures have circled Alcoholics Anonymous without meeting. The clinical literature asks whether AA works and through what mechanism; a Cochrane review concludes that twelve-step facilitation performs at least as well as comparison treatments for abstinence and operates chiefly through changes in participants’ social networks (Kelly, Humphreys, and Ferri 2020). The formal-modeling literature has produced compartmental models of drinking dynamics (Sánchez et al. 2007; Sharma and Samanta 2015), agent-based models of alcohol availability (Gorman et al. 2006), and individual-level dynamical models of behavior change fitted to clinical data (Banks et al. 2014, 2017). Neither literature has modeled the content of the Twelve Steps as a structured process, and neither has modeled the Twelve Traditions at all.

The omission is notable because the Traditions constitute an unusually clean object of institutional study. They are a written constitution for a radically decentralized organization: codified in 1946 from a decade of documented group failures, formally adopted in 1950, and unamended in substance since. They govern on the order of 120,000 autonomous groups with no enforcement mechanism, no hierarchy, and no budget above the group level. Organizations that hold a governance rule fixed for three-quarters of a century while scaling through six orders of magnitude are rare, and rarer still is one whose founding documents record the failure modes the rules were written to prevent.

The central claim is narrow, formal, and interpretive at one joint. Three of the Traditions (group conscience with servant leadership, Tradition 2; the refusal to organize hierarchically, Tradition 9; and anonymity with principles placed before personalities, Tradition 12) share an effect that is invisible when they are read as ethics and immediate when they are read as a constraint on a stochastic matrix: they prevent any member from acquiring a non-vanishing share of the group’s aggregate attention. That is precisely the condition under which DeGroot (1974) belief averaging is asymptotically wise (Golub and Jackson 2010). To our knowledge, this correspondence has not previously been stated.

The joint is worth naming before anything is built on it. The theorem is proved and was read at source. The mapping from three sentences of a fellowship’s constitution onto the theorem’s hypothesis is a reading of those sentences, arrived at by the author, and *no computation anywhere in this paper touches it*. Everything downstream inherits that status. Section 8.1 states what would settle it and Section 9 states what evidence would overturn it.

Three further contributions support the central one. We derive the coupling between Steps and Traditions from first principles, testing and rejecting the natural conjecture that the two lists pair by index, a reading the historical record already casts doubt on. We reformulate the Steps as a multistage production technology in the form of Cunha, Heckman, and Schennach (2010), under which the informal rule that steps cannot be skipped becomes the Leontief limit of a CES family and its strictness a single estimable parameter. And we build a stochastic membership model disciplined by an institutional constraint central to AA’s design: under Tradition 3, an AA group cannot refuse *membership*, so “closing the door” in the sense of excluding a person from the fellowship is not an action available to any group, and open-door adherence operates through newcomer retention rather than newcomer arrival. The constraint proves informative, separating empirically distinguishable channels of group decline.

1.1 Roadmap

Section 2 reviews the relevant literatures. Section 3 states the aggregation framework and the mapping from Traditions to its conditions, with the rotation-scaling and obstruction-scaling corollaries and the touring-speaker structure. Section 4 derives the Step–Tradition coupling and reports six robustness designs against it. Section 5 presents the CES formulation of the Steps and its identification strategy. Section 6 describes the simulation model, its results, and the sensitivity analysis that bounds them. Section 7 gives the comparative historical case. Section 8

collects limitations; Section 9 states predictions, marking where the analysis above already bears on them; Section 10 concludes. Appendix A restates the paper in plain language. Appendix B is a reproducibility note.

2. Literature Review

2.1 Effectiveness and mechanisms of AA

The Cochrane review of twelve-step facilitation (Kelly, Humphreys, and Ferri 2020) finds manualized TSF at least as effective as comparison treatments for continuous abstinence and, in several trials, superior. The mechanism literature converges on social network change: Kaskutas, Bond, and Humphreys (2002) find that AA’s effect on drinking outcomes is mediated substantially by changes in the composition of a participant’s social network; Rynes and Tonigan (2012) examine whether sponsorship effects reduce to network effects. Measurement instruments for affiliation and involvement are established (Humphreys, Kaskutas, and Weisner 1998; Tonigan, Connors, and Miller 1996; Tonigan and Greenfield 2013). The helper-therapy principle (Riessman 1965) and its AA-specific test (Pagano et al. 2004) address whether helping others benefits the helper. Galanter (1981) supplies an earlier account of why large-group affiliation relieves distress, which is the closest antecedent to this paper’s treatment of the group as a producer of resources members consume.

2.2 Formal models of drinking and recovery

Compartmental epidemic-style models treat drinking as transmissible (Sánchez et al. 2007; Sharma and Samanta 2015). Agent-based work has modeled drinking in relation to alcohol availability and outlet density (Gorman et al. 2006). Individual-level dynamical models fitted to clinical data are due to Banks and coauthors (2014, 2017). The relapse literature supplies the nonlinear, multiple-equilibrium structure this paper’s individual model borrows (Hufford et al. 2003; Witkiewitz and Marlatt 2004, 2007).

2.3 Skill formation

The Steps are treated here as a multistage technology in the sense of Cunha and Heckman (2007) and Cunha, Heckman, and Schennach (2010), with self-productivity, dynamic complementarity, and a CES aggregator whose substitution parameter carries the substantive content. Ben-Porath (1967) is the antecedent. Identification of latent skill from noisy proxies follows Schennach (2004) and Hu and Schennach (2008).

2.4 Social learning and the wisdom of groups

DeGroot (1974) gives the averaging model. Golub and Jackson (2010) give the condition under which such averaging is asymptotically wise, together with the three obstructions to it: prominent agents receiving non-vanishing attention, imbalance between attention given and received, and insufficient dispersion between subgroups. This paper’s central section is an application of that result and nothing more; the theorem is not extended.

2.5 Economics of religion, clubs, and teams

Iannaccone (1992) explains costly requirements as screening devices against free-riding in collectives; Lembke (n.d.) applies the frame to AA directly. Holmström (1982) gives the team production problem that a group producing a non-excludable good faces. Ostrom (1990) supplies the design principles for self-governing common-pool institutions, of which the Traditions are an unusually pure instance. Angrist (2014) and Carrell, Sacerdote, and West (2013) supply the

warning this paper takes most seriously about interventions built on measured peer effects; see Section 6.7.

2.6 Nineteenth-century mutual-aid temperance

The comparative case in Section 7 rests on primary temperance sources read at source rather than on the secondary literature that descends from AA’s own account: Grosh (1842), the Washingtonian movement’s own pocket manual; the autobiographies and histories of Hawkins (1862), Marsh (1866), Gough (1869), Eddy (1887), and Blair (1888); and the later scholarly treatments of Fehlandt (1904), Crothers (1911), and Krout (1925). Maxwell (1950) is the one sociological comparison of the two fellowships we have found. Kurtz (1991) is the standard scholarly history of AA and is used for the Traditions’ drafting history.

2.7 The gap

No prior work, so far as we have found, treats the Traditions as a formal constraint on an influence structure, derives the Step–Tradition coupling rather than asserting it, or models group-level membership dynamics under the institutional constraint that a group cannot refuse membership. The gap is the paper’s occasion; whether the paper fills it well is the subject of Section 8.

3. Governance as an Aggregation Mechanism

3.1 Framework

A group of N members holds beliefs about a matter of collective business. Let A be the row-stochastic trust matrix, A_{ij} the weight member i places on member j , and let beliefs update by $b(t+1) = Ab(t)$. Under strong connectivity and aperiodicity, beliefs converge to a consensus equal to $s^\top b(0)$, where s , the influence vector, is the normalized left dominant eigenvector of A . If initial beliefs are the truth μ plus independent noise of standard deviation σ , the consensus converges in probability to μ as N grows if and only if $\max_j s_j \rightarrow 0$ (Golub and Jackson 2010).

The error is available in closed form, which is how Table 3 is computed rather than simulated. The consensus is $\mu + \sum_j s_j e_j$, a normal variable with mean zero and standard deviation $\sigma \|s\|$, and for a mean-zero normal the expected absolute value is its standard deviation times $\sqrt{2/\pi}$. So

$$\mathbb{E} |\text{consensus} - \mu| = \sigma \|s\| \sqrt{2/\pi}. \quad (1)$$

Under equal weighting $s_j = 1/N$, so $\|s\| = N^{-1/2}$ and the error is exactly $\sigma \sqrt{2/\pi} / \sqrt{N}$. The single-member baseline is the same expression at $N = 1$: $\sigma \sqrt{2/\pi} = 0.798$ at $\sigma = 1$. Every figure in this section is deterministic algebra, not Monte Carlo, and carries no sampling error.

3.2 The mapping

Proposition 1 (Exchangeability under the three Traditions) *Suppose Traditions 2, 9, and 12 hold exactly, in the sense that no member is distinguished by office, by accumulated position, or by any identifying attribute on which another member could condition a trust weight. Then trust weights are exchangeable across members, A is doubly stochastic, $s_j = 1/N$ for every j , and $\max_j s_j \rightarrow 0$: the group conscience is asymptotically wise. Conversely, any arrangement assigning a fixed share $\alpha > 0$ of aggregate attention to a bounded set of members bounds $\max_j s_j$ away from zero, and aggregation fails.*

The converse is the substantive direction: violating these Traditions is sufficient to break aggregation regardless of the group’s other virtues.

Tradition (paraphrase)	Formal content	Role
2. Group conscience; leaders serve, do not govern	No row of A is privileged by office	Blocks prominence via formal authority
9. No hierarchy; service rotates	A is redrawn; office does not accumulate weight	Prevents persistent high in-degree (§3.5)
12. Anonymity; principles before personalities	Removes surname, occupation, status, the substrate of unequal weights	Blocks the formation mechanism of prominence
1. Common welfare first	A strongly connected	Precondition for convergence
4. Group autonomy	No cross-group influence aggregation	No prominent group at higher levels
3. Desire to stop drinking is the only requirement for membership	N unbounded; no screening on membership	Makes the asymptotic regime the relevant one

Table 1: The mapping. **This table is an interpretation of the Traditions’ wording, not a result.** It is the paper’s central and least verified step.

Remark 1 (The status of Proposition 1) *The mathematics is immediate; the hypothesis is the whole difficulty. Reading “leaders are trusted servants; they do not govern” as “no row of A is privileged by office” is an interpretation of a sentence, and a reader may reasonably hold that Tradition 2 is about humility rather than weighting, or that anonymity is chiefly protective of individuals and only incidentally structural. Nothing computed anywhere in this paper bears on that reading. Exchangeability is also an idealization: real trust is never exactly uniform, so the claim is asymptotic and approximate, that the Traditions push $\max_j s_j$ toward $1/N$ rather than exactly to it.*

3.3 Failure modes, quantified

Tables 2 and 3 report the two quantities separately. They are distinct and are easily conflated: at $N = 10$ the flat regime’s maximum influence weight is 0.100 and its consensus error is 0.252, and only the first is bounded by construction.

N	Flat	Dominant	Caucus of 3	Closed core of 5	Rotating, pool 12
10	0.100	0.350	0.167	0.200	n/a
50	0.020	0.350	0.167	0.200	0.041
250	0.004	0.350	0.167	0.200	0.032
500	0.002	0.350	0.167	0.200	0.030

Table 2: Maximum influence weight $\max_j s_j$. Constructions: *dominant*, one member receives 0.35 of every row, remainder split evenly; *caucus*, three members receive 0.50 of every row between them; *closed core*, five members receive 0.45 of every row and distribute their own attention only among themselves; *rotating*, one of $R = 12$ members holds share 0.35 in each term, time-averaged over the cycle (undefined at $N = 10$, where the pool exceeds the group). Exact to the digits shown; no sampling error.

N	Flat	Dominant	Caucus of 3	Closed core of 5	Rotating, pool 12
10	0.252	0.328	0.275	0.357	n/a
50	0.113	0.289	0.238	0.357	0.132
250	0.050	0.281	0.232	0.357	0.093
500	0.036	0.280	0.231	0.357	0.087

Table 3: Mean $|\text{consensus} - \text{truth}|$ at $\sigma = 1$, from equation (1), against a single-member baseline of 0.798. Same constructions as Table 2.

Under flat weighting, error declines as exactly $N^{-1/2}$; under a dominant member or an entrenched caucus it plateaus. The analytic limits confirm the tables: dominant tends to $0.35\sqrt{2/\pi} = 0.279$; the caucus tends to 0.230; the closed core sits at exactly $\sqrt{2/\pi}/\sqrt{5} = 0.357$, the error of a five-member group, because the influence of every member outside the core is identically zero at every N , which is why that column does not vary with N at all.

The closed core is qualitatively worse than the other two and we did not expect that before computing it. The first two put a floor under the group's error. The third deletes the rest of the group from the calculation: in the limit all influence accrues to the closed subgroup and the group's accuracy is exactly that of the subgroup alone, however many other people are present. This is a direct consequence of Golub and Jackson's imbalance condition and standard for absorbing sets in Markov chains; what is worth reporting is the magnitude.

Consensus is reached in every regime. Concentration does not produce visible dysfunction, only degraded accuracy delivered with undiminished confidence. Both of the first two pathologies have vernacular names in AA: the old-timer whose view settles every group conscience, and the caucus that has decided before the business meeting convenes.

Severity is not the point, and it does not take much. The illustration uses $\alpha = 0.35$, which is a great deal, and a reader could take from it that only substantial dominance matters. A member holding five per cent of every row floors the consensus error at 0.0465 against a flat 0.0252 at $N = 1000$, a factor of 1.84, and that factor grows without bound as the group grows because the flat error keeps falling and the other does not. At ten per cent the factor is 3.3.

3.4 Obstruction is a property of scaling, not of severity

Golub and Jackson's three obstructions (prominent agents, imbalance, insufficient dispersion) map naturally onto the dominant old-timer, the member whose sponsorship lineage listens to him but who listens to nobody, and the clique. **That mapping was asserted before it was computed, and computing it changed it.** The three obstructions are properties of *sequences* of societies as they grow, not properties of a room.

Hold the magnitudes fixed and neither the clique nor the imbalanced member obstructs anything. A clique of three giving a tenth of its attention outward holds, in aggregate, 0.750 of the influence at $N = 10$, 0.375 at $N = 50$, 0.107 at $N = 250$ and 0.029 at $N = 1000$. A member receiving twenty times the attention he gives holds 0.690 at $N = 10$, 0.290 at $N = 50$, 0.074 at $N = 250$ and 0.020 at $N = 1000$. Both shares vanish, both satisfy the condition, and neither is an obstruction.

Now let the same two practices scale with the group. A clique whose inwardness approaches one as the group expands holds 0.167 at every size tested from 10 to 1000. A member whose attention advantage grows in proportion to the group runs from 0.182 at $N = 10$ to 0.168 at $N = 1000$, converging to a positive share rather than settling on one exactly. Neither falls toward zero, and that is the whole of what makes them obstructions.

Corollary 2 (Obstruction scaling) *Whether a concentration of attention obstructs group learning is determined by how it scales with N , not by how severe it is at any one N . The diagnostic question about any concentration is therefore not how large it is but whether it would still be there if the group doubled.*

It is easy to read the three obstructions as three things a room can have at a given moment, and on that reading a tight clique is an obstruction wherever it appears. It is not. The general form of the point is the rotation result that follows.

3.5 Rotation must scale

Proposition 3 (Rotation breadth) *If service rotates over a pool of R members, each office-holder attracting attention share α , time-averaged maximum influence is approximately α/R*

plus residual flat weight. With R fixed as N grows, $\max_j s_j$ is bounded below and the wisdom condition fails despite rotation.

At $N = 400$ and $\alpha = 0.35$, the sweep over R gives maximum influence 0.118, 0.060, 0.031, 0.016, 0.009, 0.005 and 0.003 at $R = 3, 6, 12, 25, 50, 100$ and 400, against a flat benchmark of 0.0025. The α/R approximation tracks the computed value closely at every point, which is the check that the mechanism is the one claimed. Holding the pool at twelve while the group grows, maximum influence converges to a floor (0.041, 0.035, 0.032, 0.031, 0.030 at $N = 50, 100, 200, 400, 800$) while the flat benchmark does not (0.020, 0.010, 0.005, 0.0025, 0.0013). The ratio between them runs 2.1, 3.5, 6.4, 12.3, 23.9. That divergence is the entire result.

The pool required to come within a factor of two of flat is 13 at $N = 50$, 26 at 100, 52 at 200, 104 at 400 and 208 at 800: **twenty-six per cent of the membership throughout**, which is the operational form of the corollary.

On the size of the effect, against overstating it. A fixed pool of twelve at $N = 400$ should not be described as approaching the caucus regime. At 0.031 maximum influence and 0.089 error it sits between the flat and concentrated cases and is closer to flat on both measures, against a caucus of three at 0.167 and 0.231. It is a real but moderate permanent cost, not a failure of the same kind as a caucus.

3.6 Touring speakers: a closed form

A movement's attention structure can be modeled as bipartite: flat attention within local societies, plus cross-society attention flowing through touring speakers. Let each member give fraction *out* of their attention to the speakers and let the speakers return fraction *back* of theirs to the general membership. Then

$$\text{speakers' share of total influence} = \frac{\text{out}}{\text{out} + \text{back}}, \quad (2)$$

exact to twelve decimal places, **independent of the number of members and of the number of speakers**. Five speakers receiving three tenths of the movement's attention and returning two tenths of their own hold six tenths of the influence in a movement of ten and in a movement of a thousand alike.

This is stronger than the argument it replaces. It is not that the influence vector converges on the speakers as the movement grows; the speakers' share is fixed by a ratio and does not move with size at all, because growth adds members to the denominator of the local channel and to the numerator of the speaker channel in equal measure. Section 7 applies it.

A degenerate case, and why the parameter *back* is not decoration. With $\text{back} = 0$ the speakers form a closed communicating class, the chain is not strongly connected, Golub and Jackson's theorem does not apply, and the speakers hold *all* the influence at every size and every level of *out*, including 0.1. That is a true statement about an absorbing set and a useless one about a fellowship. The first version of this construction had that defect and reporting it would have been a vacuous-robustness error.

4. Deriving the Step–Tradition Coupling

4.1 A natural conjecture, tested

Two ordered lists of twelve, printed in a single volume, invite the conjecture that Step i draws on Tradition i . The conjecture merits testing rather than adoption. The historical record gives grounds for doubt: the Traditions were codified from 1946 essays written roughly a decade after the Steps, in response to specific organizational crises, and no AA doctrine pairs the lists by index. The stakes are concrete. Under index-pairing, Step 5 (telling one's inventory to another

person) would depend on Tradition 5 (primary purpose), though what the step plainly requires is confidentiality, which is Tradition 12; and Step 12 (carrying the message) would couple to anonymity, though Tradition 5 states nearly the same sentence. We therefore derive the coupling from first principles and test the conjecture against the result.

4.2 Method

To avoid re-deriving the same mistake, no direct step-to-tradition mapping is permitted. An intermediate layer of eight group-produced resources is introduced (admission and standing, identification, living proof, confidential audience, counsel, a recipient for twelfth-step work, continuity, and normative pressure), and the coupling is computed as

$$B = S G^\top, \quad (3)$$

where $S[i, r]$ records how much Step i 's execution consumes resource r (written by asking what each step requires) and $G[j, r]$ records how much Tradition j governs the supply of r (written independently of S). Both matrices are hand-written judgments. That is the whole of their evidentiary status, and Section 4.5 does not soften it.

4.3 Results

Table 4 gives the unperturbed coupling.

Step	Principal	Value	Runner-up	Value	Index-mate	Value	Its rank
1 admit	T3	1.22	T1	0.99	T1	0.99	2
2 believe	T11	0.82	T5	0.72	T2	0.13	7
3 decide	T2	0.31	T1	0.27	T3	0.01	7
4 inventory	T2	0.35	T1	0.35	T4	0.00	8
5 tell someone	T12	1.08	T1	0.65	T5	0.12	5
6 willing	T1	0.25	T2	0.24	T6	0.00	9
7 ask	T1	0.17	T2	0.13	T7	0.00	10
8 list harms	T2	0.43	T1	0.29	T8	0.07	4
9 amends	T2	0.90	T1	0.48	T9	0.00	11
10 daily	T1	0.94	T2	0.43	T10	0.00	12
11 connect	T1	0.47	T2	0.21	T11	0.11	4
12 carry it	T5	1.25	T3	1.10	T12	0.12	6

Table 4: The derived coupling $B = S G^\top$, unperturbed. Not one Step is chiefly served by its index-mate. Load per Tradition, exact: T1 6.52, T2 3.89, T5 3.88, T3 2.69, T11 2.69, T12 2.62, T8 1.11, and exactly zero for T4, T6, T7, T9 and T10.

1. **The two-tier division is derived, not assumed.** Five Traditions (autonomy, non-endorsement, self-support, non-hierarchy, and no outside issues) have identically zero rows in G : they govern no resource any step consumes, and can operate only by protecting the Traditions that do. This division had been conjectured informally; the derivation produces it mechanically.
2. **Unity is the most load-bearing Tradition.** Under index-pairing, unity fed one step; derived, it appears in nearly every row, because continuity and cohesion are consumed by almost everything. Tradition 1 asserts precisely this of itself; the model derives the assertion rather than encoding it.
3. **The index-pairing conjecture is rejected on all twelve counts.** Step 5's principal supplier is Tradition 12 and Step 12's is Tradition 5, the two correspondences a reader would guess, reversed.

4. **Classification is not always intuitive.** Tradition 8 (non-professionalism), which a functional reading might class as protective, lands in the enabling tier with small load, because it governs the identity of the confidential hearer, a fellow member rather than a clinician.

4.4 Six robustness designs, and what each can and cannot see

The trivial-count decomposition, stated first because it deflates the headline. T4, T6, T7, T9 and T10 have identically zero rows in G , so Steps 4, 6, 7, 9 and 10 have index-mate entries of exactly zero, and index-pairing cannot hold for them under *any* perturbation that preserves sparsity. Both multiplicative designs below preserve sparsity. The consequence is checkable and was checked: at every level, the proportion of draws in which index-pairing fails on all twelve equals the proportion in which it fails on the seven non-trivial Steps, to the last draw (98.75 and 98.75 at ± 15 per cent; 85.50 and 85.50 at ± 30 ; 72.45 and 72.45 at ± 50 ; 65.15 and 65.15 at ± 75 ; 40.60 and 40.60 structurally). **Five of the twelve counts are not evidence, and no multiplicative design could have made them so.**

Design 1, multiplicative jitter. Every entry of S and G is multiplied by an independent uniform draw on $[1 - L, 1 + L]$ for $L \in \{0.15, 0.30, 0.50, 0.75\}$; 2,000 draws per level, seed 3. This represents a reader who disagrees with the magnitudes by up to L and agrees about which cells are empty.

Design 2, structural randomization. Every non-zero entry of S and G is replaced by an independent uniform draw on $[0.05, 1.00]$; zeros stay zero. 2,000 draws, seed 23. This represents a reader who accepts only the pattern of which Tradition touches which resource and rejects every magnitude we chose. It is the same test the survival claims of Sections 6 and 7 pass at 100 per cent.

Design	T1 leads	Index-pairing wrong, all	Step 5 $\rightarrow$ T12	Step 12 $\rightarrow$ T5
jitter $\pm 15\%$	100.0 [99.8, 100.0]	98.8 [98.2, 99.2]	100.0 [99.8, 100.0]	89.7 [88.3, 91.0]
jitter $\pm 30\%$	100.0 [99.8, 100.0]	85.5 [83.9, 87.0]	99.5 [99.1, 99.7]	67.3 [65.3, 69.4]
jitter $\pm 50\%$	98.0 [97.3, 98.5]	72.5 [70.5, 74.4]	88.7 [87.2, 90.0]	52.9 [50.7, 55.1]
jitter $\pm 75\%$	86.8 [85.3, 88.3]	65.2 [63.0, 67.2]	71.5 [69.5, 73.4]	43.8 [41.6, 45.9]
structural	75.4 [73.5, 77.2]	40.6 [38.5, 42.8]	17.5 [15.9, 19.2]	27.6 [25.7, 29.7]

Table 5: Percentage of draws in which each claim holds. Wilson intervals at 95 per cent on $n = 2,000$. The two 100.0 entries are 2,000 of 2,000 and should be read as “not observed to fail”, not as certainty.

Step 12 is the fragile row and the reason is the margin, not the level: its top two are T5 at 1.25 and T3 at 1.10, and both Traditions govern the recipient resource (T5 at 0.9, T3 at 0.8), so a perturbation moving them in opposite directions flips the winner.

Design 3, the threshold test, which is the only one that can see the five. For Step i , let w_i be the row sum of S (total consumption across the eight resources) and b_i the largest entry in row i of B excluding the index-mate’s own entry. If Tradition i governed every resource at a uniform strength c , its entry in row i would be $c w_i$, so index-pairing holds at Step i exactly when

$$c > c_i^* = b_i/w_i. \quad (4)$$

This is one division per Step, carries no sampling error, and is the only design here that can turn a structural zero into a non-zero.

The governance matrix has 35 non-zero entries of 96 cells, with mean 0.374, median 0.300, minimum 0.10 and maximum 1.00. *Every* c^* exceeds both the mean and the median. The

Step	Index-mate	Beats	Row sum of S	c^*	c^* / mean live entry
1	T1	1.22	2.40	0.508	1.36
2	T2	0.82	1.70	0.482	1.29
3	T3	0.31	0.70	0.443	1.18
4	T4	0.35	0.80	0.438	1.17
5	T5	1.08	1.70	0.635	1.70
6	T6	0.25	0.60	0.417	1.11
7	T7	0.17	0.40	0.425	1.14
8	T8	0.43	0.80	0.538	1.44
9	T9	0.90	1.50	0.600	1.60
10	T10	0.94	1.50	0.627	1.67
11	T11	0.47	0.80	0.588	1.57
12	T12	1.25	2.40	0.521	1.39

Table 6: The threshold test. Bold rows are the five protective Traditions, whose index-mate entries are structurally zero and therefore invisible to Table 5. Exact; no sampling error.

protective range, 0.417 to 0.627, is not contained in the enabling range, 0.443 to 0.635; it extends below it at both ends. Mean c^* is 0.501 for the five protective Steps, 0.531 for the seven enabling ones, and 0.518 across all twelve. The protective Steps are therefore marginally *closer* to index-pairing holding, by 0.03 on the mean threshold, which is the direction an objector would predict and an order of magnitude smaller than the distance to the mean live entry. Setting c to the mean live entry of 0.374 for every Step simultaneously, all twelve index-mates still lose, by margins from 0.02 at Step 7 to 0.44 at Step 5.

What the test licenses is the statement that index-pairing would require the index-mate to govern its own Step’s needs more strongly than a typical entry in the matrix, uniformly across all twelve. What it does not license is any statement about whether the zeros are correctly *placed*: it holds the sparsity pattern’s origin fixed and prices only its consequences. It is a sensitivity analysis of a judgment, not a test of it.

Design 4, the resource-list test. Delete each of the eight resources in turn, then merge each pair, then delete each pair: sixty-four resource lists in all. Only Step 1 and Step 2 ever regain their index-mates, in fifteen variants and one respectively, and every one of the fifteen Step 1 failures involves removing or merging the admission resource, which is precisely what gives the open door its lead there. This establishes that the list is no *finer* than it needs to be. It cannot establish that the list is fine *enough*, because inventing a ninth resource requires a judgment about what it contains and cannot be done by rearranging the eight.

Design 5, the reassignment test. Kurtz (1991) records that in some later AA literature the concept conveyed by *single-purposed* was obfuscated by substituting *unity*. If the governance matrix absorbed that semantic drift, unity’s primacy is an artifact of it. For each of the eight resources in turn, set $G[T5, r]$ to the maximum of its current value and $G[T1, r]$, set $G[T1, r]$ to zero, and recompute: this transfers unity’s entire governance of one resource to singleness of purpose, which is the strongest form of the objection, and it is exact. **Six of the eight transfers leave unity leading**, by margins from 1.44 to 2.54. Transferring continuity gives T5 the lead 5.14 to 4.63; transferring pressure gives it 5.12 to 4.66. So unity’s primacy is conditional on two assignments and on nothing else in the matrix. Notably, the single resource-list variant that breaks it (Design 4) drops continuity and pressure together, which is exactly the pair this test identifies: two independent designs agreeing on which two resources carry a result is worth more than either alone. The passage Kurtz names as decisive is in an AA-copyright work this project does not acquire, so the source identified as settling the question is unread.

Design 6, sparsity perturbation. Every design above varies magnitudes and holds the sparsity pattern fixed; a second reader would disagree about the pattern. Flipping cells at random within the seven enabling rows (56 cells, so the two-tier split is held fixed), 2,000 draws per row:

Cells flipped	Index-pairing wrong, all	T1 leads	Step 5 → T12	Step 12 → T5
1	96.3%	100.0%	98.7%	94.8%
2	92.7%	99.7%	96.3%	90.5%
4	86.2%	96.5%	93.3%	80.3%
8	75.0%	83.7%	85.4%	72.0%
16	53.0%	50.8%	70.8%	53.0%

Table 7: Sparsity perturbation. Section 4 tolerates a reader differing on about four of fifty-six enabling cells and does not tolerate one differing on sixteen.

This is a bound, not a measurement, and the distinction is the caveat. A random flip is not a plausible reader. Somebody who thinks self-support governs continuity changes that cell for a reason, and their remaining cells correlate with the reason. Random flips are harsher in respecting no reason and gentler in not concentrating on the cells that carry the results. The sweep answers “how much disagreement, counted in cells” and says nothing about which cells a reader would choose. It also holds the two-tier split fixed, because the split is what defines which rows are available to flip, so the more serious disagreement, a reader who fills a protective row, cannot be represented by any perturbation design at all. Only completed elicitation forms can answer either question, and none has been returned; see Section 8.

4.5 What the coupling results rest on

Two classes of claim in this paper rest on different things and must not be defended the same way.

The **survival** claims of Sections 6 and 7 survive replacing every non-zero matrix entry with a random value. They rest on structure.

The **coupling** claims of this section do not. Index-pairing fails on all twelve in only 40.6 per cent of structurally randomized draws, which is less often than it holds; the Step 5 inversion survives in 17.5 per cent and the Step 12 inversion in 27.6. These claims rest on the magnitudes in two matrices we wrote, and they must be argued for rather than certified by a robustness percentage earned elsewhere. The one exception is unity’s primacy, which survives structural randomization in 75.4 per cent of draws, the highest figure of any claim in this section: three-quarters of the time a random matrix with the same sparsity pattern puts unity first anyway.

A reader who thinks the two matrices are arbitrary should not be persuaded by Section 4, and we would rather say so than borrow the survival claims’ robustness for a result that has not earned it.

5. The Steps as a Multistage Technology

This section is a reformulation, not the simulation’s update rule, and the two must not be read as the same object. The CES form below is a way of *stating* the ordering claim so that its strictness becomes one estimable parameter. The dynamic model of Section 6 uses the gated-growth equation given in §6.2, which is not a CES aggregator. Nothing in Section 6 depends on the value of ρ .

Let $x_{i,t}$ denote latent practice of step i . Following Cunha, Heckman, and Schennach (2010):

$$x_{i,t+1} = A_i \left[\gamma_{i1} x_{i,t}^{\rho_i} + \gamma_{i2} x_{i-1,t}^{\rho_i} + \gamma_{i3} G_{i,t}^{\rho_i} + \gamma_{i4} M_t^{\rho_i} \right]^{1/\rho_i}, \quad (5)$$

with weights summing to one, elasticity of substitution $\sigma_i = 1/(1 - \rho_i)$, $G_{i,t}$ the resource bundle from Section 4, and M_t maintenance capacity built from Steps 10 to 12. Self-productivity is $\gamma_{i1} > 0$; the cross-partial in prior-stage stock and group input is positive throughout (from +1.73 at $\rho = -4$ to +0.08 at $\rho = 0.5$), so group support is worth more to a member who has done the preceding work: dynamic complementarity in Cunha and Heckman’s sense.

Proposition 4 (The ordering rule as a limit) *As $\rho_i \rightarrow -\infty$ the aggregator converges to $\min(\cdot)$, so zero prior-stage stock forces zero output: the informal rule that steps cannot be skipped. The rule holds for all $\rho_i \leq 0$ and fails for $\rho_i > 0$, where the group input substitutes for the missing stage (output 0.217 at $\rho = 0.3$ with prior stage at zero).*

The reduction converts a widely held but untested claim of practice into a sharp empirical question: the sign of ρ decides whether the Steps are a chain or a menu, and it is estimable. CHS’s finding that substitutability falls with developmental stage yields the directional prior that ρ_i declines across the sequence; informally, several routes reach Step 3, but no substitute exists for a written inventory when amends are made.

5.1 Identification, anchoring, and design

Latent practice is observed only through noisy instruments, $Z = \mu + \lambda \ln x + \varepsilon$, with identification following Schennach (2004) and Hu and Schennach (2008) given at least three instruments per latent per wave, and the latent scale anchored in percent days abstinent at follow-up so that units are interpretable. Simulation of the estimator shows why the three-instrument requirement is substantive rather than technical: with one proxy, the estimate of ρ lands on the wrong side of zero when the truth is zero (+0.28), while three proxies recover it (+0.01), and the sign is the entire question.

The binding obstacle to estimation is endogeneity of the group input: groups direct attention to struggling members and successful members attract sponsees, so G is not exogenous to member state. CHS address the analogous endogeneity of parental investment; adapting their approach is necessary before any fit to panel data and is not attempted here.

6. Membership Dynamics Under the Open Door

6.1 The institutional constraint, restated

Tradition 3 makes the desire to stop drinking the sole requirement for **membership**. It is tempting to render this as “an AA group cannot refuse admission,” and that overstates it. AA groups hold closed meetings as a matter of routine, and a closed meeting restricts who is present in a room; it does not remove anyone’s membership in the fellowship. The constraint is on membership, not on attendance, and the model is built to that constraint. The distinction was supplied by a reader who knows the rooms, and it is the single correction this work has received that changed the most for the fewest words.

The distinction has modeling consequences. Arrival at a meeting does not depend on the group’s welcome: people come from treatment programs, courts, and desperation regardless. Staying does. Gatekeeping culture, informal requirements, and cliquishness operate on retention. Accordingly, Tradition 3 multiplies the early-tenure dropout hazard rather than the inflow; inflow contains an exogenous floor representing referral streams that arrive independent of the group’s attractiveness; and exit includes a practice-independent churn term for relocation and mortality.

Two further design choices follow from the same logic. Carrying capacity is supplied by the established core rather than the population mean, since a room of three veterans and twenty newcomers still contains three people able to carry a newcomer. And the maintenance-capacity gate, whose role is to model relapse in members with something to maintain, phases in with step index rather than throttling entry-level growth, to which it does not conceptually apply.

6.2 Model summary

Members occupy a twelve-dimensional practice state. Per-step growth, in full, for member m and step i :

$$\frac{dx_i}{dt} = h(m) a_i \text{gate}_i \text{peer}_i C_i^m (1 - x_i) - d_i x_i, \quad (6)$$

with

$$\text{gate}_i = x_{i-1}^p, \quad \text{gate}_1 = 1, \quad (7)$$

$$\text{peer}_i = (1 - \beta_i) + \beta_i G_i, \quad G = S_{\text{norm}} R, \quad (8)$$

$$C_i^m = 1 - w(i) (1 - C), \quad w(i) = 0.05 + (i - 1) \frac{0.95}{11}, \quad (9)$$

$$d_i = \delta_0 (1 + \psi(1 - x_{i+1})) \quad (i < 12), \quad d_{12} = \delta_0. \quad (10)$$

Maintenance capacity C is identical across steps for a given member and is defined in two parts. Own capacity is a Hill gate on M , the mean of that member's Steps 10 to 12, and group support supplies a floor beneath it:

$$C(M) = \frac{M^n}{k^n + M^n}, \quad n = 3.0, \quad k = 0.12; \quad C = C(M) + (1 - C(M)) \omega \bar{C}, \quad \omega = 0.75, \quad (11)$$

where $\bar{C}$ is the mean own-capacity across living members, so a member whose own maintenance has collapsed retains a fraction of capacity as long as the group around them has not. Because Steps 10 to 12 sit at the heavy end of the $w(i)$ weighting, maintenance gates its own accumulation, and the resulting loop is bistable in the individual, consistent with the relapse literature. Here $w(i)$ is per-step exposure to it, rising linearly from 0.05 at Step 1 to 1.00 at Step 12, which is the formal content of the claim that an arrival has nothing to maintain and a veteran has a great deal; and $h(m)$ is member heterogeneity, a lognormal draw made once when a member arrives and fixed thereafter, which converts individual bistability into graded group-level response.

Group resources R are produced from member states through the governance-weighted averages of Section 4, so shortfalls do not compound multiplicatively, with protective Traditions acting solely as multipliers on the enabling ones they guard. Membership is endogenous: arrivals are exogenous referrals plus attraction proportional to members' twelfth-step practice and Tradition 11; exits combine practice-dependent dropout, Tradition-3-modulated early attrition, and churn. Integration is Euler at $dt = 0.5$ weeks over a 1,560-week (thirty-year) horizon.

Parameters: $\delta_0 = 0.06$ per week (unattended half-life 11.6 weeks), $\psi = 0.20$, $p = 1.5$, step top speeds from 0.15 at Step 9 to 0.30 at Step 1. The inventory is 22 continuous scalars, 12 step speeds, 49 non-zero cells in S and 35 in the governance matrix: **118 numbers chosen by hand out of 226 cells, none fitted.**

Remark 2 (Calibration, and the exact sense in which it fails) *Inflow, dropout and churn were set so that a fully adherent group holds a steady state near forty-five members with an experienced core near nine, roughly a healthy urban meeting. At 400 seeds the model delivers 41.7 members, 95 per cent half-width 1.5, and an experienced core of 7.7, half-width 0.5, so it undershoots both, and the shortfall on the core is larger than its own interval. That is calibration to a stylized fact, not to a dataset, and it does not hit the stylized fact exactly. Note also that "core" denotes two different quantities at two thresholds: the established core, at a practice threshold of 0.1, is 37.2, and the experienced core, at 0.5, is 7.7. The two are easily conflated and are not interchangeable. Membership is reported over all runs; both core figures are conditioned on survival, which at 99.5 per cent moves nothing by more than 0.2 members.*

Remark 3 (Thirty-five parameters that cannot matter at full adherence) *The governance matrix is column-normalized, so governance quality is identically 1 when every Tradition is at 1.0 and the matrix cancels exactly. This is algebra, not simulation. All 35 governance cells*

therefore produce exactly zero change in every outcome at full adherence, and a sensitivity design that perturbs them and reports no effect has found nothing. Thirty-five of the 118 numbers are in that position.

6.3 Three channels of decline

Four configurations, 400 seeds each, thirty-year horizon, $dt = 0.5$ weeks. “Alive” means more than five members. **Quality is mean practice among members of surviving groups and is conditional throughout; the surviving fraction is printed beside it in every row.** Survival carries a 95 per cent Wilson interval; membership and quality carry a 95 per cent half-width from the cross-run standard error.

Condition	Alive y10	N if alive, y10	Quality y10	Alive y30	95% int.	Quality y30
nothing wrong	1.000 [0.990, 1.000]	46.9 ± 1.2	0.338 ± 0.0045	0.995	0.982–0.999	0.320 ± 0.0060
invisible	0.998 [0.986, 1.000]	13.9 ± 0.4	0.310 ± 0.0088	0.998	0.986–1.000	0.263 ± 0.0080
unreferred	0.907 [0.875, 0.932]	29.4 ± 1.6	0.348 ± 0.0062	0.360	0.314–0.408	0.335 ± 0.0099
unwelcoming	0.993 [0.978, 0.997]	31.8 ± 1.4	0.357 ± 0.0059	0.940	0.912–0.959	0.335 ± 0.0084

Table 8: Decline scenarios, **400 seeds**, 1,560-week horizon. The seed count is not incidental: at ten seeds this table returns a referral-loss survival near 0.20 and a full-adherence membership near 45.0, both outside the intervals reported here. *Invisible* sets Tradition 11 to zero; *unreferred* sets exogenous inflow to zero; *unwelcoming* sets Tradition 3 to zero. The year-thirty quality figures for the unreferred and unwelcoming rows are identical to three decimals and their intervals overlap almost entirely, so they should be read as indistinguishable rather than as equal.

Membership counted over all runs with deaths as zero reaches, at year thirty, 41.7 ± 1.5 , 13.5 ± 0.4 , 9.9 ± 1.6 and 27.5 ± 1.7 respectively.

Three signatures separate. A group that becomes invisible to seekers shrinks to a stable remnant sustained by referrals and does not die. A group whose referral stream dries up survives on attraction for a decade and then dissolves in 64 per cent of runs, while per-member quality among survivors holds essentially to the end. A group with exclusionary internal culture shrinks by roughly a third and keeps its core: a size effect rather than a survival effect.

The mortality profiles follow from structure rather than from hand-tuning. A group on one engine has no floor underneath it: its inflow becomes a function of its own state, which makes the population dynamics multiplicative rather than additive, and a multiplicative process with no floor has an absorbing state at zero. Invisible and unwelcoming groups are smaller but retain a floor, so they persist. Unreferred groups are the only ones whose inflow can go to zero and stay there.

The quality claim, stated exactly. At year ten, the measure a member could plausibly perceive, quality against a healthy group is 0.028 *lower* for the invisible condition, 0.010 *higher* for the unreferred one, and 0.019 *higher* for the unwelcoming one. Only the first has the sign an observer would expect. The distinctive prediction is therefore not that quality is uniformly maintained but that *in the condition that kills groups, quality does not fall and slightly rises*, so the interior of a dying group offers no warning.

6.4 The gap between decline and death

The unreferred condition separates two questions that a single membership series conflates.

This is why the survival-versus-size distinction is load-bearing throughout the paper. A group that is dying and a group that is small look the same in an unconditional average.

6.5 Comparative statics under variance control

Discipline preceded comparison: baseline survival is stable across independent seed blocks, and common random numbers reduce the paired standard error of membership comparisons roughly

Series	y5	y10	y20	y30
unreferred, all runs	33.4 ± 1.2	27.0 ± 1.7	16.0 ± 1.7	9.9 ± 1.6
unreferred, survivors only	33.5	29.4 ± 1.6	25.8	26.6 ± 3.0
surviving fraction	0.995	0.907 [0.875, 0.932]	0.603	0.360 [0.314, 0.408]

Table 9: Decline versus death under referral loss. The unconditional series falls by seventy per cent; the conditional series falls by twenty. The gap between them is mortality and nothing else.

fourfold relative to the cross-seed standard deviation of 11.93. Each Tradition is degraded singly from 0.85 to 0.5 against a reference group of 23.44 members.

Tradition	Tier	Members cost	95% half-width	t
T11 attraction	enabling	7.90	1.20	12.9
T3 open door	enabling	5.16	1.25	8.1
T1 unity	enabling	4.47	1.30	6.7
T2 group conscience	enabling	3.43	1.30	5.2
T4 autonomy	protective	2.82	1.37	4.0
T7 self-support	protective	2.82	1.37	4.0
T12 anonymity	enabling	2.56	1.24	4.1
T5 one purpose	enabling	2.27	1.33	3.4
T8 non-professional	enabling	1.02	1.20	1.7
T9 no organization	protective	0.89	1.15	1.5
T6 no endorsement	protective	0.71	1.13	1.2
T10 no outside opinion	protective	0.71	1.13	1.2

Table 10: Single-Tradition degradation, **400 paired replications** under common random numbers. Eight of twelve reach conventional significance.

The replication budget determines the ranking, and 30 is not enough. Computed at 30 paired replications, this comparison returns autonomy and self-support at 5.5 members each with $t = 2.6$ as the only pair clearing $|t| > 2.5$, which invites the inference that the protective Traditions lead the ranking, consistent with their derived role as guards on everything else. At 400 replications they sit fifth and sixth at 2.82 each, attraction leads by a factor of nearly three, and eight comparisons resolve rather than two. **A common-random-numbers design at 30 replications is more efficient than 30 independent runs and is still 30 replications.** Variance reduction buys precision per replication; it does not substitute for replications, and any ranking reported at that budget is an artifact of it.

6.6 Sensitivity, and the two results that survive it

The simulation’s parameters were subjected to: global multiplicative jitter; a tiered design; 236 one-at-a-time targeted perturbations covering all 118 hand-chosen values; Morris elementary-effects screening; Sobol total-order indices; four structural variants that change the model’s architecture rather than its numbers; and a resource-list test. Full specifications are in the book’s technical appendix, sections A3 to A9.

Which parameters matter. The ordering exponent p has the largest single influence on group maintenance of any parameter, swinging it from 0.4386 to 0.0045 against a baseline of 0.1458, a range of 2.98 times baseline; the decay rate is second at 1.83 and member heterogeneity third at 1.69. Sobol total-order indices on membership agree: p at 0.456 [0.331, 0.599], δ_0 at 0.258, heterogeneity standard deviation at 0.178, dropout at 0.134 and churn at 0.121, against a noise floor of 0.023 to 0.034 (membership and practice respectively). **The first-order indices were computed and are not usable**, and no number from that column is quoted anywhere. Three diagnostics establish it: estimated S_1 exceeds S_T for p on membership and for five of eight

factors on practice, which violates the identity $S_T \geq S_1$ that holds for any true decomposition; the practice first-order indices sum to 1.263, whereas a sum of first-order indices cannot exceed one; and the estimates move substantially with the seed block. Resolving a first-order index to a half-width of 0.10 needs a base sample roughly ten times larger, so the first-order decomposition of a stochastic model of this size is out of reach at the budget available. The total-order indices are reported because they are stable; the interaction structure is not identified. The sum of S_T is 1.300 for membership, and the excess over one is the signature of interaction counted once per factor involved, so interactions are present and are not dominant.

Structural variants. Four changes to the architecture, not the numbers: a flat per-step gate, admission moved to Tradition 3, capacity supplied by all members rather than the core, and a clipped saturation term. Under the attraction-loss condition, survival runs 0.998 (base), 0.825, 0.998, 1.000, 0.998 and membership 13.5, 8.4, 13.5, 14.7, 14.0. Under referral loss, survival runs 0.360, 0.490, 0.360, 0.635, 0.573 and membership 9.9, 15.6, 9.9, 20.3, 21.7.

What survives all of it, and in exactly which sense. Two simulation results survive the whole suite. The first is that a fully adherent group persists. The second is that losing referrals is worse than losing attraction *in the sense of being more likely to kill the group*: it holds under all four structural variants and by a wide margin, and it survives all 236 targeted perturbations, which makes it the strongest thing the simulation says. Read instead as “ends up smaller,” the same ordering *reverses* under three of the four structural variants: in the gate-flat, capacity-all and no-saturation variants the referral-loss condition ends larger than the attraction-loss one. **The claim must not be written without saying which reading is meant.** Specific survival figures survive nothing, and every one in this paper is a conditional statement about a model.

What the designs could in principle have found. Two failures are worth naming because both were made in this project. A multiplicative perturbation cannot move a structural zero, so no multiplicative design is evidence about the two-tier split (Section 4.4). And a ± 30 per cent jitter cannot tell you whether a result depends on the magnitudes at all, only whether it tolerates small disagreement about them; the structural randomization is the design that answers the first question, and it is the one the coupling claims fail.

6.7 Founding composition: a null, and why it is weak

Twenty-five founders with a fixed total practice of 13.75 distributed three ways (even: all at 0.55; concentrated: five at 1.00 and twenty at 0.4375; split: twelve at 0.90 and thirteen at 0.2269), 400 seeds each. Membership ends at 41.7, 41.3 and 39.9 with half-widths 1.5, 1.5 and 1.6; survival at 0.995, 1.000 and 0.998; quality among established members at 0.3539, 0.3534 and 0.3514. The whole spread across conditions is 1.8 members against an uncertainty of 1.6. The split condition is lowest on all three measures, which is the direction Carrell, Sacerdote, and West (2013) would predict, and it is not significant and we do not claim it.

The null should be discounted heavily before it is read. Composition can enter this model by exactly two routes: the non-linear capacity gate and member heterogeneity. Resources are computed from sums and means over the living membership, and no member’s state appears in another member’s growth equation except through those aggregates, so the model has no representation of mentoring, of pairing, of cliques, or of sponsorship in particular. **It also has no representation of the mechanism that produced the Carrell–Sacerdote–West result**, which is people choosing whom to associate with inside a group whose composition has been arranged. That experiment engineered Air Force Academy squadrons from measured peer effects, predicted a gain of 0.053 grade points for the bottom third, and measured a treatment effect of -0.061 on exactly the students it set out to help, because the low-ability cadets re-sorted

toward each other. A null from a design that could barely have produced anything else is the sort of result that looks like evidence and is not.

The design also cannot separate two things: the conditions differ in the variance of founding practice *and* in the number of founders above the established-member threshold (25, 5 and 12). Had a difference appeared, it could not have been attributed to dispersion rather than to initial core size.

7. The Comparative Case

A formal correspondence gains little from a single historical case and can lose a great deal by leaning on one. This section is included because the case bears directly on the paper's central claim, and because the standard account of it, which is what a reader is most likely to bring, turns out to be false.

7.1 The received account, and what the primary source says

The Washingtonian Total Abstinence Society was founded in Baltimore in April 1840 by six working men, in a scene whose earliest surviving account is Harrison (1860) and which reaches most modern readers through Maxwell (1950), spread nationally within four years on claims reaching into the hundreds of thousands, and was effectively finished within a decade. In the literature descending from AA it is standardly described as a movement that died of having no rules, and it is used to illustrate why the Traditions matter. That literature arrives with its moral pre-attached.

The movement's own manual falsifies the account. Grosh (1842), the *Washingtonian Pocket Companion*, printed at Utica and by then in a second edition, carries in its first fifteen pages a definition of principles, directions for organizing and conducting meetings, and a model constitution. Among its provisions: each society independent and subordinate to none; funds controlled by its own members; and nothing sectarian or political admitted to lectures, speeches, singing, or the doings of the society. A footnote records that a Washingtonian mass convention at Utica passed a declaration of principles and a model constitution on 22 February 1842, that it was printed in the *Utica Washingtonian* of 25 February and reprinted in October because of demand, and that a copy should be procured wherever a society is organized. **They had written rules, they had them within two years of founding, and they had a mechanism for transmitting them.**

What they had, in writing, were analogues of four of the Traditions this analysis classifies as protective. What they had none of were the seven classified as enabling. And on anonymity they took the opposite position deliberately and with an argument: Grosh's directions for a first meeting have joiners rise and call out their names for the secretary, because "publicity and freedom are preferable to private solicitations, whisperings, and secresy in giving the names."

The comparison the case actually supports is therefore not rules against no rules. It is one written code against another, and the provisions missing from the first are the ones the aggregation condition points at.

7.2 Prominence without anonymity

The touring-speaker structure of §3.6 was built for this case. Washingtonian expansion from 1841 to 1843 proceeded through touring speakers addressing local societies, which is exactly a rise in the proportion of total attention carried by a one-directional cross-society channel. By equation (2) the speakers' share of influence is fixed by the ratio of outward to returned attention and does not fall as the movement grows, so the movement could not have outgrown the exposure.

The exposure was realized. John B. Gough, the movement’s most prominent speaker, relapsed publicly in September 1845; opponents seized on it, and public confidence in the movement was impaired. His own account, read at source, contains the words “I have fallen,” an acceptance of blame, and a submission to his church’s judgment. It is worth reading rather than inferring from its chapter headings, which suggest a man rebutting a charge rather than owning a relapse. He had relapsed once before, twenty-nine months earlier, when his influence weight was small, and that episode was handled internally and quickly. Same man, same illness, same candor: what differed was the weight.

7.3 Maxwell, and a priority problem stated plainly

Milton Maxwell’s 1950 comparison of the two fellowships reaches, without any formal apparatus, a substantial part of this paper’s conclusion. His final section lists AA’s advantages as exclusively alcoholic membership, singleness of purpose, a definite program of recovery, anonymity, and what he calls hazard-avoiding traditions; he writes that a comparison with the Washingtonian experience underscores *the sheer survival value* of the principle of anonymity; and he reaches that conclusion by way of Gough’s relapse and what it cost a movement whose credibility sat in named men. He also identifies the tradition of keeping authority in principles rather than in offices and personalities, and connects it to rotating leadership.

What this paper adds to Maxwell is the theorem, and nothing else. He had the observation, the mechanism, and the case. He had no formal condition to which the observation could be referred, and therefore no way to say why anonymity should have survival value rather than merely that it did. Whether the vanishing-influence condition is what Maxwell was pointing at is a separate question, and this paper does not settle it.

7.4 What the case cannot do

It is one case, selected because it is the obvious comparison, and the direction of selection is unfavorable: the Washingtonians are famous among people interested in AA precisely because the contrast is instructive. Nothing here establishes that the missing enabling provisions caused the decline; the movement was also absorbed by a temperance politics it had defined itself against, and the manual’s anti-politics article had nothing to bite on once the movement’s identity was itself a political position. How widely the Utica model constitution was actually adopted is not recorded in the manual and we have found no source that settles it. The historical material is offered as an existence proof that the distinction between the two tiers of rule is visible in a real code, not as evidence about the consequences of omitting one tier.

8. Limitations

8.1 The largest one, stated first

The mapping in Table 1 is an interpretation of the Traditions’ wording, arrived at by the author, and no computation in this paper touches it. Everything the paper claims about AA specifically, as opposed to about stochastic matrices, passes through it. A reader who holds that Tradition 2 concerns humility rather than weighting, or that anonymity is chiefly protective of individuals, can accept every number here and reject the paper’s thesis. This is not a caveat on a result; it is the status of the result.

What would settle it is elicitation: give the Traditions’ published short text to readers who do not know the hypothesis, ask them to say what each rule constrains, and measure agreement with the mapping. That has not been done. A single reader who knows the rooms did read a draft and identified a substantive error, the membership-versus-attendance conflation of §6.1,

and no other intervention in the project changed as much for as few words. The implication for the parts nobody has checked is uncomfortable and is the reason this limitation is listed first.

8.2 The remaining limitations

1. **No parameter is estimated.** Section 3’s computations demonstrate a theorem on constructed matrices; Section 6’s are simulations from 118 assumed values, none fitted, because the longitudinal data such a model would need has never been collected. The paper’s empirical content is its predictions, not its numbers. A model of this kind can show that a set of ideas is consistent and that a mechanism is available. It cannot show that anything is true.
2. **The S and G matrices are hand-written judgments,** and all of Section 4 plus the resource structure of Section 6 inherits from them. Their tolerance of ± 30 per cent perturbation is necessary and not sufficient support, and the structural randomization they fail (§4.5) is the design that speaks to the question. The correct remedy is a second governance matrix elicited independently from another reader, scored by Cohen’s kappa against the first. A blank elicitation form exists; no second reader has completed it. Until one does, Section 4 rests on one person’s judgment.
3. **Member heterogeneity was tuned to produce a desired behavior.** Its standard deviation was selected because it converts individually bistable dynamics into graded group response. The mechanism is principled; the magnitude is not, it is the third most influential parameter in the model, and Section 6’s quantities inherit from it.
4. **Specification search.** Several architectural choices (the retention channel for Tradition 3, core-based carrying capacity, the placement of the capacity gate) were refined against simulation behavior as well as substantive reasoning. This risks tailoring an architecture to expected behavior and no assurance can be given that it has not happened. The four structural variants of §6.6 are a partial and inadequate answer.
5. **Thirty-five parameters cannot affect a fully adherent group at all,** because the governance matrix cancels exactly at full adherence. Any sensitivity result quoted over the full 118 must be read with that in mind, and a design that perturbs those 35 and reports no effect has found nothing.
6. **Conditional figures read as unconditional.** Quality in Table 8 is computed among members of surviving groups. Read without the surviving fraction beside it, “quality is maintained” would describe a population that has partly ceased to exist. Table 9 exists to make the difference visible.
7. **DeGroot averaging is a strong simplification of a group conscience.** Real members argue, defer selectively, abstain, update out of order, and sometimes harden rather than converge. Golub and Jackson’s result is about naive averaging specifically. Whether real deliberating groups behave like DeGroot updaters has not been tested here or, so far as we have found, anywhere.
8. **Endogeneity of the group input** blocks estimation of Section 5’s technology and is unsolved here.
9. **The sign of ρ has never been measured.** The contribution is to make the question answerable, not to answer it.
10. **No representation of who attends to whom.** In Section 6’s model no member’s state enters another member’s growth equation except through group aggregates, so there is no sponsorship, no pairing, no clique, and no re-sorting. The mechanism that produced the Carrell–Sacerdote–West result is absent by construction (§6.7). Building it is a different model, not a different run, and it is the largest single piece of technical work outstanding.
11. **Durations were never tested.** The horizon, the arrival rate and the churn floor are

three of the 118 hand-chosen numbers. The *orderings* in Table 8 survived the sensitivity work; the *durations* did not and should be read as shapes, not as forecasts of how long a real group lasts.

12. **Combinations were not tested.** Each decline condition switches one thing off at full adherence elsewhere. Real decline is unlikely to be so tidy and there is no reason to expect the costs to add.
13. **A correspondence is not a cause.** That the Traditions implement a known aggregation criterion does not establish that this is why they were written, or why AA has endured.
14. **Sources not read.** Alexander (1988) on the class and domestic dimensions of the Washingtonian movement; Blumberg (1980, 1991) on its political entanglement; the full text of Pagano et al. (2004), which is cited at a remove; White's *Slaying the Dragon*. Maxwell (1950) was read in full, but the copy available to us is a retyped web reproduction with visible transcription errors, not a scan of the journal, so every Maxwell citation here has been checked against a transcription rather than against the journal. AA's own literature was deliberately not acquired.

9. Falsifiable Predictions

Where the analysis reported above bears on a prediction, or narrows what it should say, this is marked.

1. **Decision quality improves with group size under flat influence and plateaus under concentrated influence.** Testable with forecast or vignette tasks administered to group consciences of varying structure. Unchanged, and untested.
2. **Rotation breadth, not rotation per se, predicts group durability.** Testable from service rosters. *Narrowed by the analysis above:* the operational threshold is roughly a quarter of the membership in the rotation pool, computed as twenty-six per cent across sizes from 50 to 800 (§3.5). The effect of falling short is a real but moderate permanent cost, not a near-clique regime, and a test should be powered accordingly.
3. **Groups losing exogenous newcomer inflow decline demographically while per-member engagement among remaining members is maintained.** Unlisted meetings, schedule-disadvantaged meetings, and meetings distant from referral sources provide natural variation. Quality-maintained decline distinguishes this model from demoralization accounts, which predict that quality and size fall together. *Narrowed by the analysis above:* the prediction is not that quality is uniformly maintained across all decline modes. In the invisible condition quality falls; in the referral-loss condition it rises slightly. The testable claim is that *the decline mode that kills groups is the one with no interior warning signal*.
4. **The two supply channels dissociate:** attraction loss produces stable remnants, referral loss produces delayed dissolution, so proximity to treatment facilities should predict survival through low-attraction periods. *Partly answered, and qualified.* This is one of exactly two simulation results that survive the full sensitivity suite, and it survives *as a claim about mortality*. As a claim about eventual size it reverses under three of four architectural variants. Any test must be a test of survival, not of membership.
5. **Gatekeeping culture predicts group size, not group survival.** *Bearing of the analysis above:* at 400 seeds, unwelcoming groups end at 27.5 members against 41.7 with 94.0 per cent surviving against 99.5, so the effect on survival is small and non-zero rather than absent. The prediction should be stated as predominantly a size effect.
6. **If step order does not predict step completion** ($\rho > 0$ throughout), the chain structure fails and the Steps model should be discarded rather than repaired. Unchanged, and the estimation design of §5.1 requires at least three instruments per latent per wave.

7. **the two-tier division should be visible to readers who do not know the hypothesis.** Give the published short text of the Traditions to independent readers and ask which of them supply anything a member uses directly and which only protect other rules. If the derived split (Section 4.3) does not recover from independent judgment, the resource layer is encoding the author’s prior rather than the text’s content.
8. **the mapping itself is testable by elicitation.** See §8.1. This is the prediction whose failure would cost the paper the most.

10. Conclusion

This paper’s durable contribution is a correspondence: three of AA’s Twelve Traditions, on a reading of their wording that the paper states plainly and does not verify, jointly implement the vanishing-influence condition under which naive collective deliberation aggregates information, a criterion formalized by Golub and Jackson in 2010 and reached by a fellowship of laypeople, from eleven years of watching groups fail, in 1946. The correspondence yields two corollaries with immediate empirical content, that rotation must scale with the group and that obstruction is a matter of scaling rather than severity, and it reframes anonymity from an ethic of humility to a structural precondition of trustworthy group decision-making.

Around that center, the paper contributes a method: deriving institutional couplings through an explicit resource layer rather than asserting them, which rejects the natural index-pairing conjecture on all twelve of its instances while recovering, unbidden, the functional division of the Traditions and the primacy of unity. It also contributes an accounting of what that method cannot support. Five of the twelve rejections are structurally forced and invisible to the perturbation design that was originally quoted for them; a threshold test that can see them finds them failing by comparable margins; and the central coupling claim survives structural randomization less often than it fails. Those facts are reported here because a derivation whose robustness is asserted rather than measured is worth less than one whose limits are known.

The membership model’s principal prediction, that referral-starved groups decline demographically while looking healthy from inside, is specific, mechanistically grounded, distinguishable from rival accounts, and testable with records AA’s service structure already keeps. It is also one of only two simulation results that survive sensitivity analysis over all 118 parameters and four architectural variants, and it survives as a claim about mortality rather than about size.

Where this line of work should go next is not further modeling. It is measurement: of the mapping, by elicitation from readers who do not know the hypothesis; of the governance matrix, by a second independent elicitation; of rotation breadth against group longevity; of step ordering against step completion; and of the quiet demographic signature of groups that seekers have stopped finding.

References

- Alexander, R. M. (1988). “‘We Are Engaged as a Band of Sisters’: Class and Domesticity in the Washingtonian Temperance Movement, 1840-1850.” *Journal of American History* 75(3): 763-785.
Not read; cited for the women’s dimension of the movement, which this paper does not develop.
- Angrist, J. D. (2014). “The perils of peer effects.” *Labour Economics* 30: 98-108.
- Banks, H. T., K. L. Rehm, K. L. Sutton, C. Davis, L. Hail, A. Kuerbis, and J. Morgenstern (2014). “Dynamic modeling of behavior change.” *Quarterly of Applied Mathematics* 72: 209-251.
- Banks, H. T., K. Bekele-Maxwell, R. A. Everett, L. Stephenson, S. Shao, and J. Morgenstern (2017). “Dynamic modeling of problem drinkers undergoing behavioral treatment.” *Bulletin of Mathematical Biology* 79: 1254-1273.
- Ben-Porath, Y. (1967). “The production of human capital and the life cycle of earnings.” *Journal of Political Economy* 75(4): 352-365.

- Blair, H. W. (1888). *The Temperance Movement: or, The Conflict Between Man and Alcohol*. Boston: William E. Smythe. **Read at source**; public domain.
- Blumberg, L. U. (1980). "The Significance of the Alcohol Prohibitionists for the Washingtonian Temperance Societies." *Journal of Studies on Alcohol* 41(1): 37-77. **Not read**.
- Carrell, S. E., B. I. Sacerdote, and J. E. West (2013). "From Natural Variation to Optimal Policy? The Importance of Endogenous Peer Group Formation." *Econometrica* 81(3): 855-882. doi:10.3982/ECTA10168. **Read at source** from the lead author's university copy. Earlier circulated as NBER Working Paper 16865 and, before that, as *Beware of Economists Bearing Reduced Forms?*. In copyright; full text not redistributed.
- Crothers, T. D. (1911). *Inebriety: A Clinical Treatise*. Cincinnati: Harvey Publishing. **Read at source**; public domain.
- Cunha, F., and J. J. Heckman (2007). "The technology of skill formation." *American Economic Review* 97(2): 31-47.
- Cunha, F., J. J. Heckman, and S. M. Schennach (2010). "Estimating the technology of cognitive and noncognitive skill formation." *Econometrica* 78(3): 883-931. **Read at source**.
- DeGroot, M. H. (1974). "Reaching a consensus." *Journal of the American Statistical Association* 69(345): 118-121. **Read at source**; the updating model.
- Eddy, R. (1887). *Alcohol in History*. New York: National Temperance Society. **Read at source**; public domain.
- Fehlandt, A. F. (1904). *A Century of Drink Reform in the United States*. Cincinnati: Jennings and Graham. **Read at source**; public domain.
- Galanter, M. (1981). "The 'relief effect': A sociobiological model for neurotic distress and large-group therapy." *American Journal of Psychiatry* 138(5): 588-591.
- Golub, B., and M. O. Jackson (2010). "Naïve Learning in Social Networks and the Wisdom of Crowds." *American Economic Journal: Microeconomics* 2(1): 112-149. **Read at source**. The wisdom criterion, the three obstructions, and the convergence conditions. Every formal claim in Section 3 originates here.
- Gorman, D. M., J. Mezic, I. Mezic, and P. J. Gruenewald (2006). "Agent-based modeling of drinking behavior." *American Journal of Public Health* 96(11): 2055-2060.
- Gough, J. B. (1869). *Autobiography and Personal Recollections of John B. Gough*. Springfield, Mass.: Bill, Nichols & Co. **Read at source**; public domain. The September 1845 episode in his own words.
- Grosh, A. B., comp. (1842). *Washingtonian Pocket Companion*. Second edition. Utica, N.Y.: B. S. Merrell. **Read at source**; Harvard copy digitized by Google, via HathiTrust, <https://hdl.handle.net/2027/hvd.32044004487591>. Public domain. The definition of principles, the model constitution's articles, the Utica mass convention of 22 February 1842, and the directions for taking names publicly.
- Harrison, J. F. (1860). *A Voice from the Washingtonian Home*. Boston. **Read at source**; public domain. The earliest account of the founding scene.
- Hawkins, W. G., ed. (1862). *Life of John H. W. Hawkins*. Boston: Briggs and Richards, sixth thousand. **Read at source**; public domain. Note the edition: this is not the Jewett printing usually cited.
- Holmström, B. (1982). "Moral hazard in teams." *Bell Journal of Economics* 13(2): 324-340.
- Hu, Y., and S. M. Schennach (2008). "Instrumental variable treatment of nonclassical measurement error models." *Econometrica* 76(1): 195-216.
- Hufford, M. R., K. Witkiewitz, A. L. Shields, S. Kodya, and J. C. Caruso (2003). "Relapse as a nonlinear dynamic system." *Journal of Abnormal Psychology* 112(2): 219-227.
- Humphreys, K., L. A. Kaskutas, and C. Weisner (1998). "The Alcoholics Anonymous Affiliation Scale." *Alcoholism: Clinical and Experimental Research* 22(5): 974-978.

- Iannaccone, L. R. (1992). "Sacrifice and stigma: Reducing free-riding in cults, communes, and other collectives." *Journal of Political Economy* 100(2): 271-291.
- Kaskutas, L. A., J. Bond, and K. Humphreys (2002). "Social networks as mediators of the effect of Alcoholics Anonymous." *Addiction* 97(7): 891-900.
- Kelly, J. F., K. Humphreys, and M. Ferri (2020). "Alcoholics Anonymous and other 12-step programs for alcohol use disorder." *Cochrane Database of Systematic Reviews*, CD012880.
- Krout, J. A. (1925). *The Origins of Prohibition*. New York: Alfred A. Knopf. **Read at source**; public domain. Independent corroboration of the founding, the officers, the fee and the dues.
- Kurtz, E. (1991). *Not-God: A History of Alcoholics Anonymous*. Expanded edition. Center City, Minn.: Hazelden. **Consulted at source**; in copyright, full text not stored. The Traditions' drafting history. A vocabulary-only verification index is retained at `research/kurtz-1991-verification-index.json` in place of the text.
- Lembke, A. (n.d.). "Sacrifice, stigma, and free-riding in Alcoholics Anonymous." Association for the Study of Religion, Economics and Culture. **Cited at a remove**.
- Marsh, J. (1866). *Temperance Recollections*. New York: Charles Scribner. **Read at source**; public domain.
- Maxwell, M. A. (1950). "The Washingtonian Movement." *Quarterly Journal of Studies on Alcohol* 11: 410-452. **Read in full**, with a caution: the available copy is a retyped web reproduction carrying visible transcription errors, not a scan of the journal. Every citation to Maxwell in this paper has been checked against that transcription rather than against the journal.
- Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge: Cambridge University Press.
- Pagano, M. E., K. B. Friend, J. S. Tonigan, and R. L. Stout (2004). "Helping other alcoholics in Alcoholics Anonymous and drinking outcomes." *Journal of Studies on Alcohol* 65(6): 766-773. **Cited at a remove**; the full text could not be obtained.
- Riessman, F. (1965). "The 'helper' therapy principle." *Social Work* 10(2): 27-32.
- Rynes, K. N., and J. S. Tonigan (2012). "Do social networks explain 12-step sponsorship effects?" *Psychology of Addictive Behaviors* 26(3): 432-439.
- Sánchez, F., X. Wang, C. Castillo-Chávez, D. M. Gorman, and P. J. Gruenewald (2007). "Drinking as an epidemic." In K. Witkiewitz and G. A. Marlatt (eds.), *Therapist's Guide to Evidence-Based Relapse Prevention*, 353-368.
- Schennach, S. M. (2004). "Estimation of nonlinear models with measurement error." *Econometrica* 72(1): 33-75.
- Sharma, S., and G. P. Samanta (2015). "Analysis of a drinking epidemic model." *International Journal of Dynamics and Control* 3: 288-305.
- Tonigan, J. S., G. J. Connors, and W. R. Miller (1996). "The Alcoholics Anonymous Involvement (AAI) scale." *Psychology of Addictive Behaviors* 10: 75-80.
- Tonigan, J. S., and B. L. Greenfield (2013). "The general Alcoholics Anonymous tools of recovery." *Psychology of Addictive Behaviors* 27(3).
- Witkiewitz, K., and G. A. Marlatt (2004). "Relapse prevention for alcohol and drug problems." *American Psychologist* 59(4): 224-235.
- Witkiewitz, K., and G. A. Marlatt (2007). "Modeling the complexity of post-treatment drinking." *Clinical Psychology Review* 27(6): 724-738.

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of *Age, Pass It On, Twelve Steps and Twelve Traditions*, the *Grapevine* essays of 1946, and service pamphlets) were deliberately not acquired for this project.

What was not read. Any work testing whether real deliberating groups behave like DeGroot updaters. Any literature on peer-group composition in voluntary mutual-aid settings specifically. Alexander (1988), Blumberg (1980, 1991), Pagano et al. (2004) in full, and White’s *Slaying the Dragon*. Jellinek’s per-capita consumption estimates, which are quoted through Maxwell and whose original has not been traced.

A. The Whole Paper in Plain Language

This appendix says everything the paper says, without the math. It is written for a reader with no background in economics or statistics. Where a natural way of putting something would be wrong, this appendix says so.

A.1 What this paper is about

AA runs on two sets of twelve ideas. The Twelve Steps are for the person. The Twelve Traditions are for the group. This paper turns both into math and checks whether the pieces fit together, and along the way it finds that some things everybody assumes about them are wrong, and some things nobody says about them are true.

A.2 The big finding: why “nobody’s in charge” actually works

Think about how an AA group makes a decision. There is no boss. People talk it over until the room agrees. That is called the group conscience.

Now here is a question: when can you trust a decision made that way? In 2010, two economists proved the answer. A group that decides by talking it out can be trusted only if no single person’s opinion carries a big fixed chunk of the final answer. Here is why. Everyone’s opinion is partly right and partly mistaken. When lots of opinions get blended evenly, the mistakes point in different directions and cancel out, and the more people the better. But if one person’s opinion always makes up a third of the result, their mistakes never cancel. Adding more people does not help. The group just gets more and more sure of an answer that is no more likely to be right.

Now look at three of the Traditions. Leaders serve the group; they do not run it. There is no ladder to climb, because service jobs rotate. And everyone is anonymous: no last names, no job titles, no status. Each of those rules does the same thing from a different angle: it stops any one person’s voice from getting too heavy. Which is exactly the condition in the 2010 proof.

So the AA members who wrote those rules in 1946, after eleven years of watching groups fall apart, landed on a rule that mathematicians would not prove correct for another 64 years. They did not have the math. They had the wreckage.

One honest warning about that paragraph, and it is the most important sentence in this appendix. The proof is real and we have read it. But the step from “Tradition 2 says leaders do not govern” to “therefore nobody’s opinion carries extra weight in the math” is our reading of a sentence. Nobody has checked it. None of the computer work in this paper tests it. Somebody could reasonably say those Traditions are about humility, not about arithmetic, and this paper has no answer for them yet. Everything else here is downstream of that one unverified step.

A.3 What it looks like when the rule breaks

Two familiar characters break it. The first is the old-timer whose opinion ends every discussion, not because there is a vote, but because everyone waits to hear what they think. The second is the little circle of long-timers who have already talked it over before the meeting starts.

Here is the alarming part from the math: when this happens, nothing looks wrong. The group still reaches agreement. The meetings feel fine. There is no argument and no drama. The group is simply wrong more often, it is wrong confidently, and getting bigger does not fix it. A broken group conscience feels exactly like a working one from the inside.

There is a third case, worse than both, that we did not expect. If a small group of people not only receives most of the attention but gives all of *their* attention only to each other, then in the long run the rest of the room counts for literally nothing. The group's judgment becomes exactly the judgment of that handful, no matter how many other people are in the chairs.

And it does not take a monster to cause the problem. One person holding five per cent of the room's attention, which is not very much, already nearly doubles how wrong a large group ends up.

A.4 Something the Traditions forgot to mention

The Traditions say to rotate service jobs so nobody becomes a boss. The math agrees, and adds a catch nobody wrote down. Rotating only works if lots of different people take turns. Picture a group of 400 where the same twelve people trade the jobs among themselves forever. Rotation like that leaves a permanent floor under how much attention the rotating dozen hold, and the floor does not go down as the group grows. To really work, roughly a quarter of the group needs to be in the rotation. Any group could check this against its own service list.

One thing not to overstate. It would be easy to say that a group of 400 rotating twelve people behaves almost the same as a group run by a clique. That is too strong. It sits in between, and it is closer to the healthy case than to the clique. The honest description is a real but moderate permanent cost.

And a bigger lesson underneath it. The obvious question to ask about a prominent member is "how prominent?" That is the wrong question. It is "does the prominence grow when the group grows?" A clique that keeps to itself in a room of ten but gets diluted as the room fills is not a problem at all in the long run. A single person who stays equally prominent no matter how big the room gets is a problem forever. Severity is not the test. Scaling is the test.

A.5 The steps don't match the traditions by number

It is tempting to think Step 1 goes with Tradition 1, Step 2 with Tradition 2, and so on down the line. It sounds neat. But it does not hold up, and it was never AA teaching. The Traditions were written about ten years after the Steps, to fix real problems groups were having.

Instead of matching them by number, we asked a different question for each step: what does a person actually need from the group to do this step? Then the matching takes care of itself. Step 5 is telling another human being the worst things you have ever done. What do you need for that? You need to know it will not leave the room. That is anonymity, Tradition 12, not Tradition 5. Step 12 is carrying the message to the alcoholic who still suffers. Tradition 5 says the group's one job is to carry its message to the alcoholic who still suffers. Same sentence, basically. Those two go together.

And when you do this for all twelve steps, two things fall out. First, five of the Traditions (autonomy, no endorsements, paying your own way, no hierarchy, no outside opinions) turn out to give members nothing directly. Their whole job is to protect the other seven from being eroded. The Traditions have two different kinds of rule in them, and nobody says so. Second, unity shows up as necessary for nearly every step, not just the first one. Tradition 1 literally says that, and it always sounded like a nice slogan. The math backs it as a plain fact.

Now the part that argues against the finding. We tested how much of this survives if you disagree with the numbers we made up.

Unity's importance holds up well. Even if you throw away every number we chose and keep only the pattern of which rule touches which need, unity still comes out on top about three times in four.

The rest does not hold up nearly as well. Do the same thing to the "steps-don't-match-traditions" finding and it only survives about four times in ten, which is less often than it fails. Worse, five of our twelve pieces of evidence for it were not really evidence at all: those five Traditions had zeros in our table by construction, so of course they never matched. We built a different test that could actually see those five, and they still fail, by about the same margins as the other seven. So we think the finding is right. But it rests on our judgment about the numbers, not on a proof, and anyone who thinks we chose the numbers badly is entitled to reject it. That is worth saying plainly rather than dressing it up.

A.6 Can you skip a step? Nobody has ever checked

Everyone in AA says you cannot skip a step, that you cannot make honest amends (Step 9) for harms you never wrote down (Step 4). It sounds obviously true. But as far as we could find, no researcher has ever actually tested whether people work the steps in order, or whether doing an early step really is what makes a later one possible.

This paper turns that belief into one measurable number. If the number comes out one way, the steps are truly a chain: skip a link and everything after it fails. If it comes out the other way, they are more like a menu, and a strong group can carry someone past a step they missed. The number could be measured with the questionnaires researchers already use, with one warning from our tests: you need at least three different measurements per person, because with only one the answer can come out backwards.

A.7 What can actually go wrong for a group

Start with a fact about AA that shapes everything here: the only requirement for *membership* is a desire to stop drinking, and no group can take that away from anybody.

One thing that is easy to get wrong here. It is tempting to put this as "a group cannot close its doors," and that is not right. Closed meetings are ordinary and long-standing, and a closed meeting is about who is in the room, not about who is a member. Nobody can be thrown out of AA. Somebody can absolutely be in a room where they are not welcome. Getting that distinction right is what moves the question from who arrives to who comes back, which is where the interesting answers turn out to be. We owe the point to a reader who knows the rooms.

So if a group cannot revoke anyone's membership, what can actually go wrong? Three different things, and they leave three different marks:

1. **The group becomes invisible.** Nobody finds it anymore: it is not listed, it meets at a bad time, word of mouth dried up. The model says this group shrinks to a small huddle kept alive by court cards and treatment-center referrals. It does not die. It just gets small, and the people in it are somewhat worse off than they would be in a healthy group.
2. **The referrals dry up.** The treatment center closed; the court program ended. The group coasts on word of mouth for about ten years, and then, in roughly two runs out of three, it quietly dissolves.
3. **The group gets unwelcoming.** Nobody loses their membership, but there is an in-crowd, and newcomers can feel it. They come once and do not come back. This group shrinks by about a third and mostly survives. It is smaller than it should be.

How much you can trust that number depends entirely on how many times the simulation was run. From ten runs, that middle case looks like four groups in five dissolving.

From four hundred, it is closer to two in three, and a healthy group settles at about 42 members rather than 45. Ten runs is far too few. A number produced by a random simulation and quoted to three digits from ten runs is not a result at all, and it is very easy to print one without noticing.

And here is the strangest thing the model says. In the case that actually kills groups, the people still in it are doing fine. Their recovery holds up right to the end, and if anything looks slightly *better* than average, because the ones who were struggling have already gone. From the inside, that dying group looks like a healthy group. Nobody in the room feels anything going wrong. If that is true in the real world, and it is checkable, it means a group cannot rely on how the meetings feel to know whether it is in trouble. It has to look at the numbers: how many newcomers came this year, and how many came back.

One qualification, because the loose version of this is wrong. Quality does not hold up in all three cases. In the invisible case it drops noticeably. The sharp version of the claim is narrower and more useful: *the failure that kills groups is the one with no warning signal inside the room.*

A.8 Does it matter who starts a group?

We tested it: send twenty-five founders into a new group, either all about the same, or a few very experienced ones plus a lot of beginners, or half strong and half weak. After thirty years, all three come out the same. Nothing.

You should not believe that result very much, and the reason is a famous experiment. Researchers at the Air Force Academy measured how much cadets help each other study, then used those measurements to build squadrons designed to help the weakest students. Their own model predicted a gain. What actually happened was a loss of about the same size, on exactly the students they had tried to help. The reason was that once you put fifteen strong students and fifteen weak ones in a room with nobody in between, the weak ones stopped mixing with the strong ones and found each other instead. The measurement was real; building a room out of it destroyed the thing that had been measured.

Our model has no way for people to choose who they spend time with. So it could never have produced that result, and its “nothing happens” answer is nearly worthless as evidence. What is worth saying is the combination: engineering who is in the room has a small upside if any, and a downside that has actually been measured once.

A.9 The movement that came before, and what it actually did

AA was not the first fellowship of drunks helping each other stay sober. In 1840, six men in a Baltimore tavern founded the Washingtonians. It spread across the country in four years and was essentially finished in ten.

The story usually told, especially in AA circles, is that they died of having no rules. **We checked, and that story is false.** Their own handbook from 1842 turned up, and it contains written rules: each society independent and answering to nobody above it, funds controlled by its own members, and nothing political or religious allowed into the meetings. They passed a model constitution at a convention in February 1842, printed it in a newspaper, reprinted it because of demand, and told every new society to get a copy.

What they did not have was any of the rules on the other side of the ledger, the ones that give a member something. And on anonymity they took the opposite view on purpose, with a reason: their handbook tells new joiners to stand up and call out their names, because publicity is better than “whisperings and secrecy.”

So the real comparison is not rules against no rules. It is one written code against another, and the ones missing from the first are exactly the ones the math points at. That is a better comparison than the one we started with, and it is better because it is true.

Their most famous speaker relapsed in public in 1845, opponents made the most of it, and the movement’s credibility suffered for it. He had relapsed once before, quietly, two and a half years earlier, when hardly anyone knew his name, and that time it cost nothing. Same man. Different weight.

A.10 How much of this should you believe?

Here is the honest answer, and it is less flattering than the summary above may have sounded.

The mathematical theorem in A.2 is real and settled. The claim that AA’s three rules are the same thing as the theorem’s condition is *our reading of three sentences*, and nobody has checked it. That is the whole hinge of the paper and it is the part with the least support.

The finding about rotation follows from the theorem and is solid math.

The step-and-tradition results come from two tables of numbers we made up, and they do not survive if you throw those numbers away. We say so in the paper rather than quoting a robustness figure that belongs to a different claim.

Everything from the computer model has never been checked against real people or real groups. Every number in it was chosen by us, not measured. When we tested how much of it survives disagreement about those numbers, exactly two things survived: that a group following all the rules keeps going, and that losing referrals is more likely to kill a group than losing walk-ins. Even that second one only holds if you mean “more likely to die.” If you mean “ends up smaller,” it flips around when we change how the model is built.

A model like that can show that ideas fit together. It cannot show they are true.

Treat the group-decision math as settled mathematics, treat its application to AA as an unverified reading, and treat everything else as sharp questions waiting for someone to check them. Be suspicious of any idea, ours included, that feels certain before it has been tested. The claims here that felt most obvious at the outset are, without exception, the ones that needed the most narrowing.

One thing this paper must never be used for: judging any individual person’s recovery. It has nothing to say about whether you, or anyone, is “doing it right.” It has never been tested on a single human being. For anything personal, talk to a sponsor, a doctor, or a counselor.

A.11 The one-sentence version

The rule that nobody in AA is in charge and nobody uses their last name may not be just humility: read a certain way, it is the exact mathematical condition that makes a group’s decisions trustworthy, written down by people who learned it from failure sixty-four years before anyone proved it. Whether that reading is right is the thing still to be checked.

B. Reproducibility Note

Every quantitative claim in this paper is reproducible from the companion repository without reference to any external dataset, because there is no external dataset.

Deterministic results. Sections 3 and 4 are algebra on constructed matrices. Influence vectors are normalized left dominant eigenvectors; consensus errors come from the closed form in equation (1); the coupling is a matrix product; the threshold test is one division per Step. These carry no sampling error and are exact to the digits printed.

Stochastic results. Section 6 reports Monte Carlo output. The design is: Euler integration at $dt = 0.5$ weeks over 1,560 weeks; **400 seeds minimum for any published figure**; survival reported with Wilson intervals; membership and quality with 95 per cent half-widths from the cross-run standard error; Tradition comparisons under common random numbers at 400 paired replications. **Common random numbers reduce variance and do not reduce the replication count required:** the comparison in Table 10 was originally run at 30 paired replications, produced a different ranking, and reversed when rerun at 400.

Perturbation designs. Multiplicative jitter at $L \in \{0.15, 0.30, 0.50, 0.75\}$, 2,000 draws per level, seed 3. Structural randomization replacing every non-zero entry with a uniform draw on $[0.05, 1.00]$, 2,000 draws, seed 23. Wilson intervals throughout on $n = 2,000$. A multiplicative design cannot move a structural zero and is not evidence about sparsity; the structural design is not evidence about where the zeros belong.

Named threats. Selection effects in the historical case (§7.4); conditioning on survival, addressed by printing the surviving fraction beside every conditional quality figure; specification search on architecture (§8.2); the exact cancellation of 35 governance parameters at full adherence; and, largest of all, the unverified mapping of §3.2.

Verification. All figures are asserted against recomputed values in a notebook executed end to end in a single process, which exits non-zero on any exception, any failed assertion, or any cell producing no output. A separate checker verifies consistency *between* documents: that every decimal is reachable from the notebook, that simulation output states its seeds and an interval, and that no citation attributes to a source something that source does not contain. Each of those checks exists because an error of exactly that kind got as far as a written page before it was caught.